

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		13.44/13.4/13 = 3 marks If incorrect award 1 mark for each of breathing rate = $5/25 \times 60 = 12$ (1) volume of CO ₂ = $5.6/5 = 1.12$ (1)		3		3	3	3
		(ii)		colour change is {subjective/qualitative}/ref. to difficulty of noting when colour has changed			1	1		1
	(b)	(i)		more {energy/ATP} is needed (for exercise) (1) from <u>aerobic</u> respiration (1)		2		2		
		(ii)		(more {energy/ATP} is now being released by) <u>anaerobic</u> respiration (1) Lactic acid production (causing cramp) (1)		2		2		
		(iii)		Individual 4 because they had <u>cramp</u> {at a lower intensity of exercise/ soonest}			1	1		
	(c)			more accurate measurement of aerobic respiration/ref. validity of conclusions/increased confidence			1	1		
				Question 7 total	0	7	3	10	3	4